

CSE7000 - INTERNSHIP

A Presentation on

F.R.I.D.A.Y. - A PERSONAL VOICE ASSISTANT CHATBOT

Student Details	
Name	Reiyan Tahleel
Roll No	20241CSE1010
Section	5CSE15

Under the **Supervision** of,

Mr. Sakthi S and Ms. Shaik Salma Begum
Professor / Associate Professor / Assistant Professor
Presidency School of Computer Science and
Engineering
Presidency University

Name of the Program: B. Tech (CSE)

Name of the HoD: Dr Asif Mohamed

Name of the Program Internship Coordinator: Dr. R. Balakrishnan

Name of the School Internship Coordinators: Dr. Bhuvaneshwari Patil



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About Company or Organization

- AI Centre of Excellence (Accelerated by NVIDIA)
- Industry-oriented training initiative focused on Artificial Intelligence and modern computing.
- Provides hands-on learning in AI, Machine Learning, Deep Learning, Generative AI, and Full-Stack Web Development.
- Offers practical exposure through labs, projects, and real-world applications.
- Aims to equip students with industry-ready technical and problem-solving skills.



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Working domain or the technology

- Working Domains

Artificial Intelligence

(AI) Deep Learning

- Large Language Models

(LLMs) GPU Computing

- Full-Stack Web
Development

Technologies & Tools

- Python
- Google Antigravity
- NVIDIA H200 Tensor Core GPU
- CUDA
- HTML, CSS, JavaScript
- Ollama

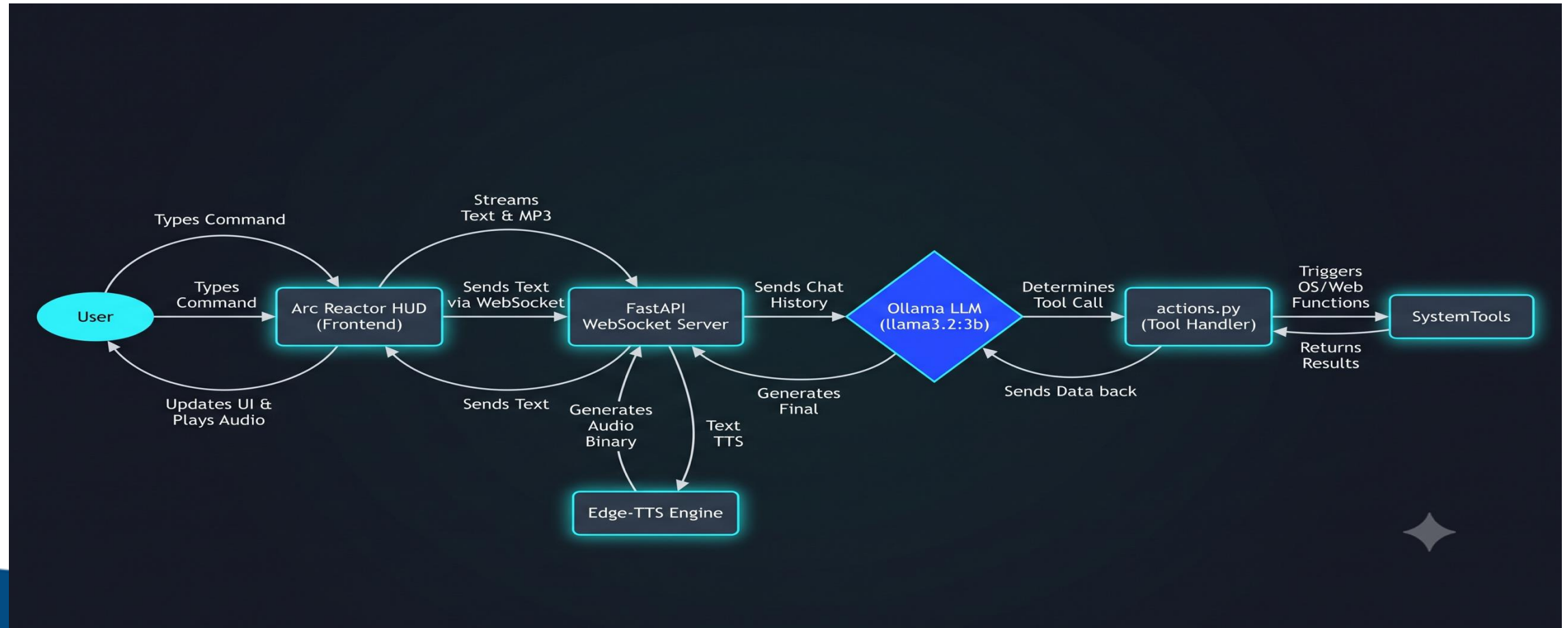


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Work flow



Objectives of the work

- **Privacy-First AI:** To implement a virtual assistant that processes requests locally using Llama 3.2, minimizing reliance on cloud APIs.
- **Task Automation:** To automate routine system tasks (e.g., opening applications, playing music on YouTube, math calculations) via natural language.
- **Low-Latency Interaction:** To create a seamless, real-time interface using Web Sockets between an Iron Man-inspired web HUD and the Python backend.
- **Scalability:** To design a modular tool-calling architecture where new capabilities (Python functions) can be dynamically added to the AI's repertoire over time



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Methodology Used / Course Description

- **Client-Server Architecture:**
- **Backend:** Python Fast API handling WebSocket connections.
- **Frontend:** HTML/CSS/JS rendering the interactive Arc Reactor HUD.
- **AI Tool Calling (Function Calling):** The SLM is provided with a strict set of tools calculate math, search web, play music, open application. It dynamically decides which Python function to trigger based on the prompt.



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- **Action Execution Map:** A modular actions.py script that maps SLM decisions to system commands using libraries like os, pywhatkit and duckduckgo search.
- **Dynamic Audio Streaming:** Generating fast audio responses via edge tts and instantly streaming the binary audio files to the client
- The Python backend passes the user's input to the locally hosted Ollama model (llama3.2:3b).
- The LLM acts as the brain, analysing the intent and determining if a specific "tool" needs to be called based on its system prompt.



Results

- **Project Outcomes & Interface**
- The AI assistant successfully processes natural language and executes system commands with high accuracy and low latency.
- The use of local LLMs ensured that conversational context was maintained securely.

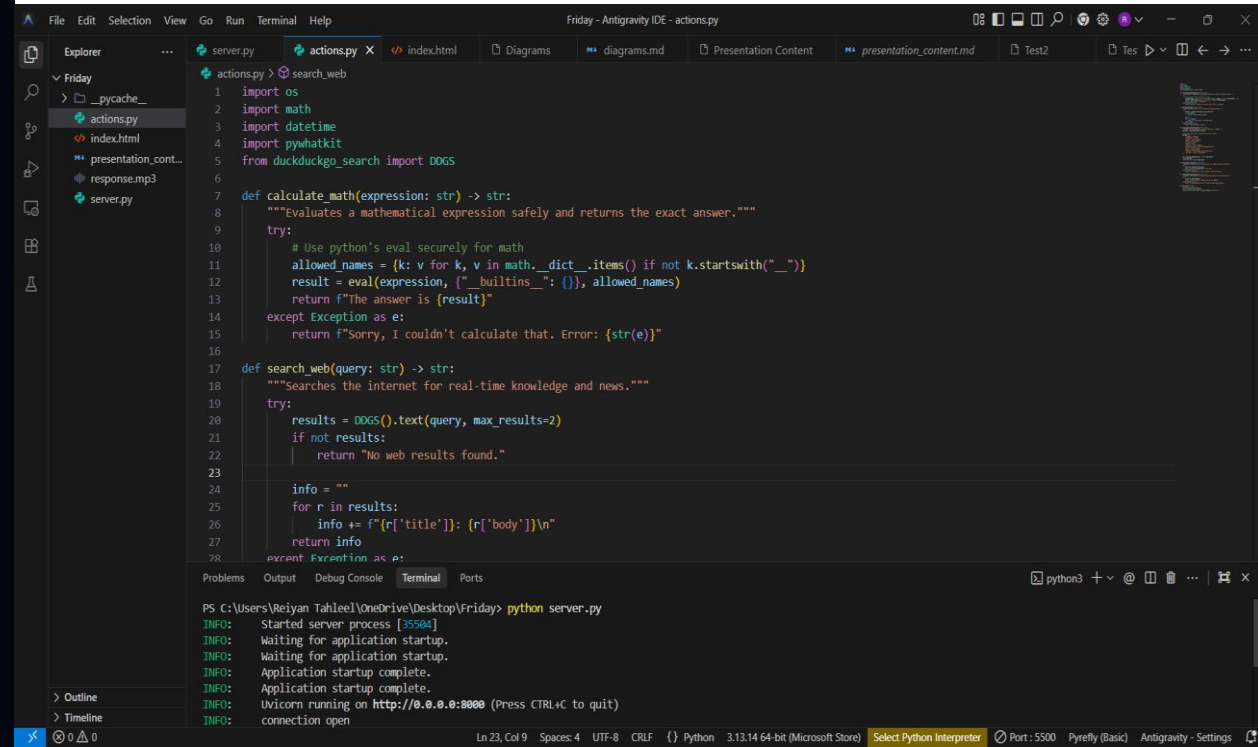
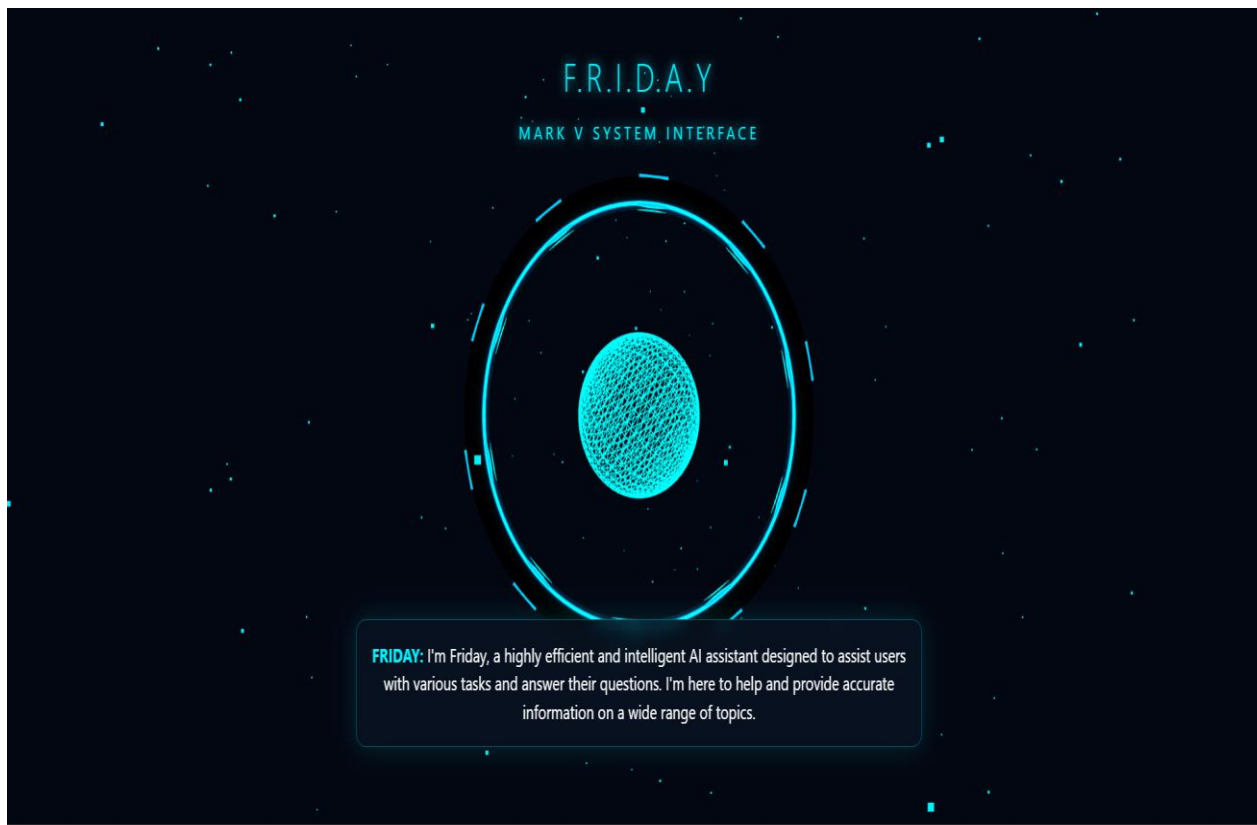


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Screenshots



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Conclusion

- **Summary :** The internship provided hands-on experience in building a complex, SLM AI application . The "Friday" assistant successfully met its objective of combining an immersive UI with powerful local SLM-driven automation.
- **Key Learnings:** Mastery of ,Integrating local LLMs (Ollama) with tool calling, and Training LLMs



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Q&A



Thank you !!



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